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LAB SESSION 1

1. Advantages of UML

It is an internationally recognized standard language for designing the software system diagrams. Providing a way to visualize and understand process flows before the development (coding) stage to reduce mistakes. It helps to Bridge the gap between technical people i.e developers and non - technical stakeholders and provide comprehensive documentation of the software system. UML is not only use for software systems, but also non - software systems.

2. Explain different kinds of UML diagram and their significance.

UML is majorly divided into two parts; Structural diagrams and Behavioural diagrams.

Structural Diagrams

1. Class diagram: It helps in coding the structure of the software. It shows classes, properties, attributes and their interactions with each other. Class diagram models object - oriented programming.
2. Component diagram: It shows how modules / components are connected together. Smaller units/components are connected to make a larger unit. It shows the static aspect of the system.
3. Object diagram: It is linked with the objects of a class and their relationship with each other along with their attributes and methods.
4. Package diagram: It is a type of diagram that depicts the organization of various elements within a software system.
5. Deployment diagram: It shows the software mapping on hardware nodes.

Behavioral Diagrams

1. Use case diagram: It shows how various users interact with different functionalities. It depicts the interaction between user and system.
2. Sequence diagram: Shows the sequence of interactions between objects.
3. Activity diagram: Shows workflows and step-by-step activity processes.
4. Communication diagram: Shows how different objects interact by passing messages.
5. Statechart Diagram: Shows the possible states of the diagram as well as the events that trigger those states, providing a comprehensive overview of the system's states.

3. Explain some possible challenges of UML in software development in your own words.

1. If any team member interprets diagrams differently, it may cause confusion.
2. UML is not always necessary for every project, specifically small projects.
3. Special tools are required to make UML diagrams, which may be cost effective.
4. Many kinds of UML diagram for a single software may lead to increased complexity.
5. Spending more time on UML may lead to delayed development.

4. Real world examples of UML

E-commerce, Healthcare, Social media management, Hostel management, Banking Systems.

5. State 4+1 view Model and why it is called 4+1 instead of 5 View model

The 4+1 view model describes the architecture of the system from perspectives of different stakeholders.

It has four main viewpoints:

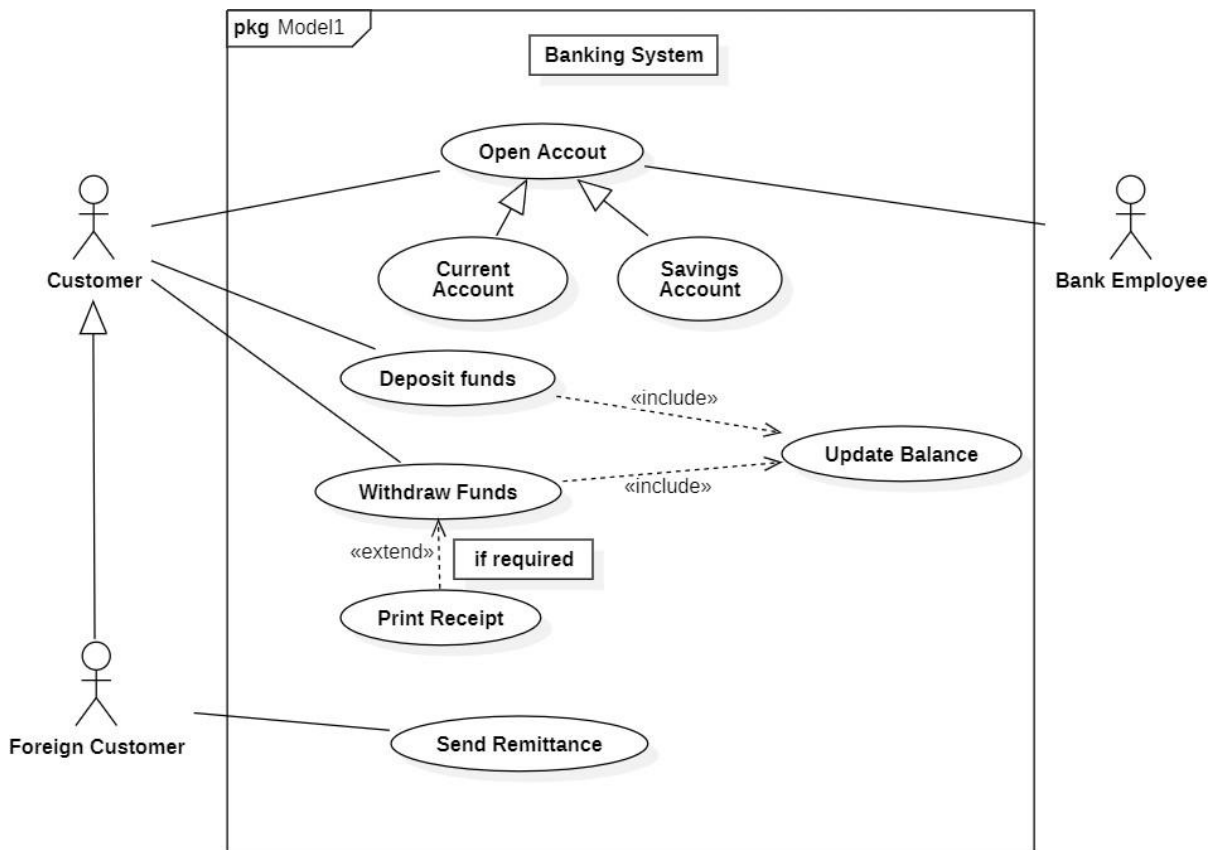
1. Logical: describe what the system does?
2. Process: describe how the system performs and runtime behaviour.
3. Development: Discuss the modularity of the system.
4. Deployment: discuss where the software runs on hardware.
5. The +1 view: Use-case/Scenario view
 - Joins all the views together to address all stakeholders at a single place.
 - It's not an independent view but works as a validator to ensure connectivity of other four views.

LAB SESSION 2

1. Install Staruml and state its key features in your own words

1. User - friendly interface
2. Flexible to model multiple kinds of diagrams
3. Can run on multiple platforms (macOS, Windows etc.)
4. Diagrams support PNG, PDF formats etc.

2. Create a dummy project and insert any UML diagram and set properties through Editor Holders



3. Compare StarUML with any other similar tool and state their significance and limitations.

Limitations of other tools:

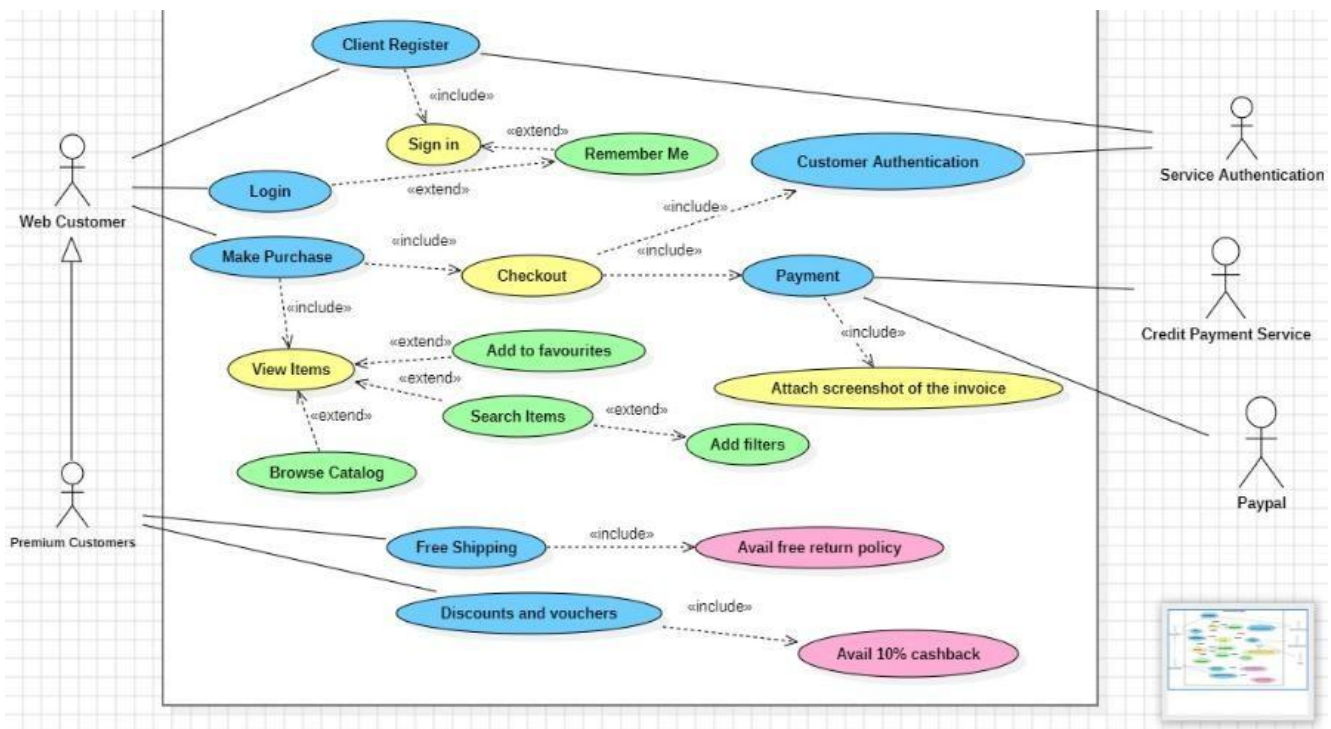
1. Cost for licensing can be higher, especially for small and medium scale projects.
2. The free version lacks advanced features like code generation.
3. Collaboration in free version is limited.
4. Performance degradation as compared to StarUML
5. Free version may not have enough storage etc.

StarUML is a powerful, lightweight software with advanced tools and offline working support. Moreover, it is flexible for individual developers and small teams.

LAB SESSION 3

1. Draw a Use-Case diagram for an Online Shopping System:

Hint: Web Customer actor uses some web site to make purchases online. Use cases are View Items, Make Purchase and Client Register. View Items use case could be used by customer if he only wants to find and see some products. This use case could also be used as a part of Make Purchase use-case. Client Register use case allows customer to register on the web site through <<Service>>Authentication actor. Note, that Checkout use case is included use case not available by itself - checkout is part of making purchase. View Items use case is extended by several optional use cases - customer may search for items and browse catalog. Checkout use case includes Customer Authentication. It could be done through user login page (sign in), Remember me page. Actor: Service Authentication Checkout use case also includes Payment use case which could be done either by using credit card and external credit payment service or with PayPal. (Actors: Credit Payment Service, PayPal).

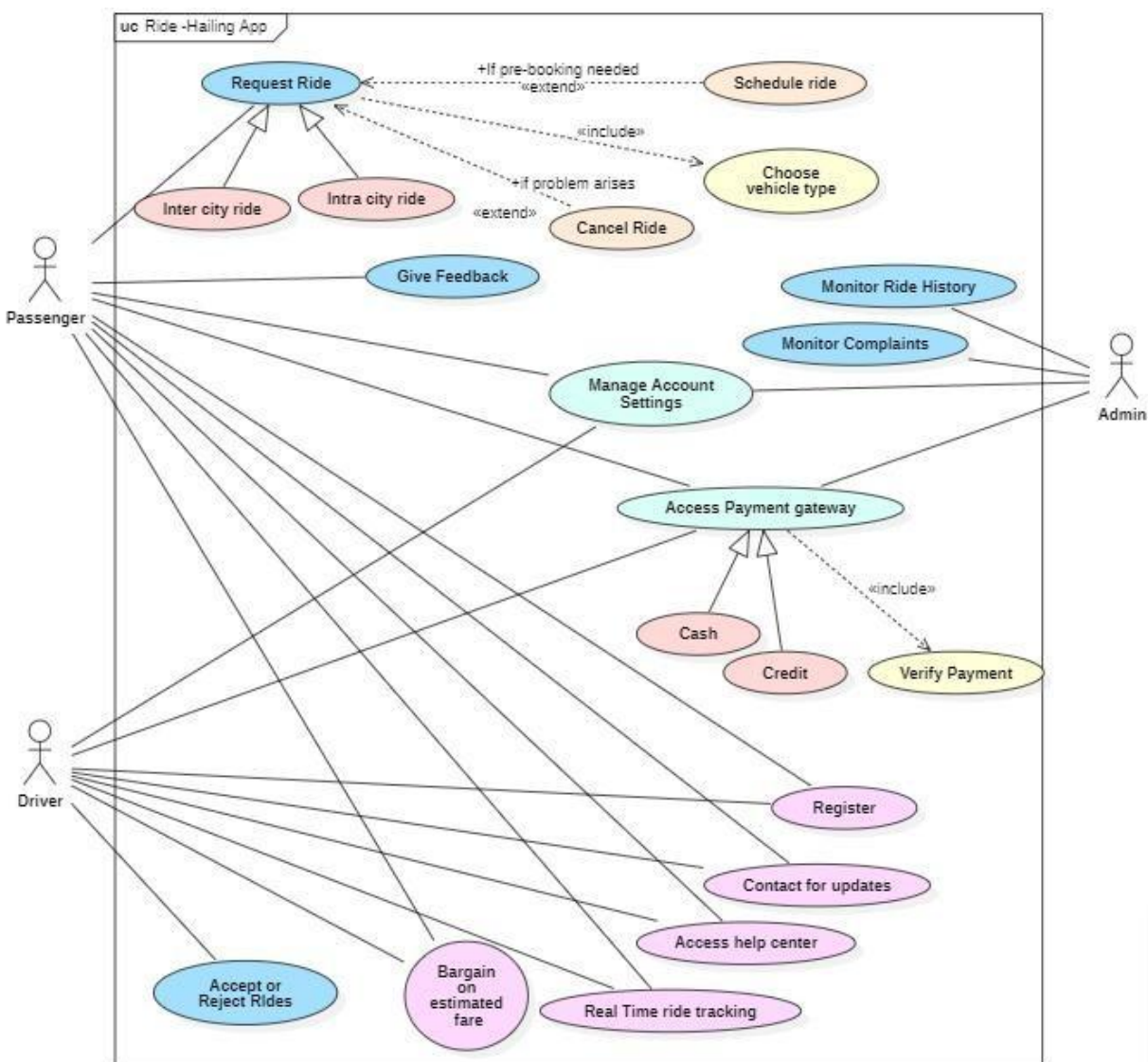


2. A city-based ride-hailing platform allows passengers to request rides using a mobile app.

Drivers can accept or reject ride requests, navigate to pick-up points, and drop passengers at their destinations. The system supports real-time tracking, fare estimation, and multiple payment options (cash, credit card, wallet). Admins manage accounts, handle complaints, and monitor ride history. A payment gateway processes all digital payments.

- Identify Actors – List all users and external systems interacting with the application.
- Identify Use Cases – List the main actions each actor can perform.
- Draw the Use Case Diagram in StarUML(attach Printout)
- Show the system boundary as Ride-Hailing App.
- Place actors outside and use cases inside.
- Use include and extend relationships where applicable.
- Export your diagram as PNG and submit both the .mdj file and image file.

Hint: Consider features like ride booking, cancellation, ride tracking, payments, and ratings.



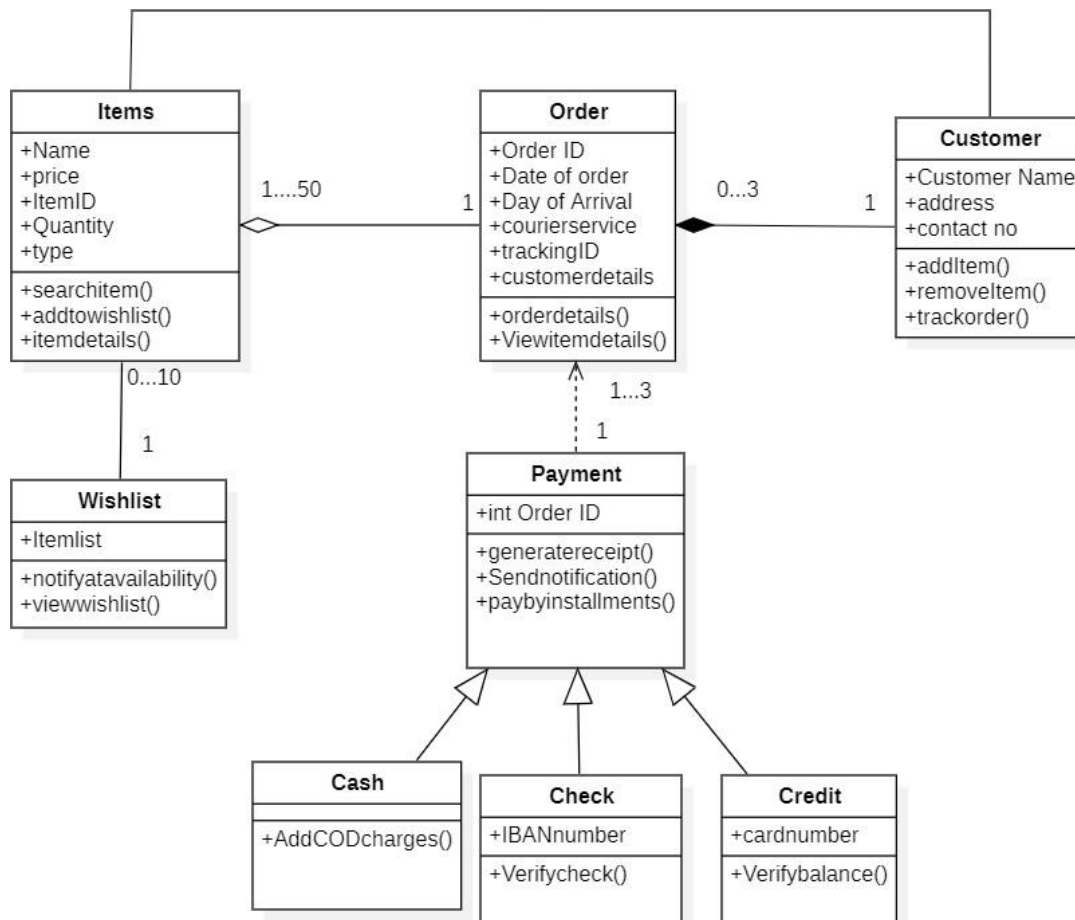
LAB SESSION 4

1. Draw a class diagram for Retail Catalog Order:

Hint: The central class is the Order. Associated with it is the Customer making the purchase and the Payment. A Payment is one of three kinds: Cash, Check, or Credit. The Order contains OrderDetails, each with its associated Item. Note: Make necessary assumptions and Show relationship among classes.

Solution: For question 1, we assumed the following:

- We assumed that the customers HAD orders, and without the customers the order would be destroyed. As such - composition
- We assumed the items would still exist even if the order was destroyed as such- aggregation
- We assumed payment was dependent on the order because without an order there would be no payment
- We assumed customers could pay in installments, as such, 1 order can have up to 3 payments
- We assumed the amounts (one to many) of orders, customers etc.
- we assumed a Wishlist exists.

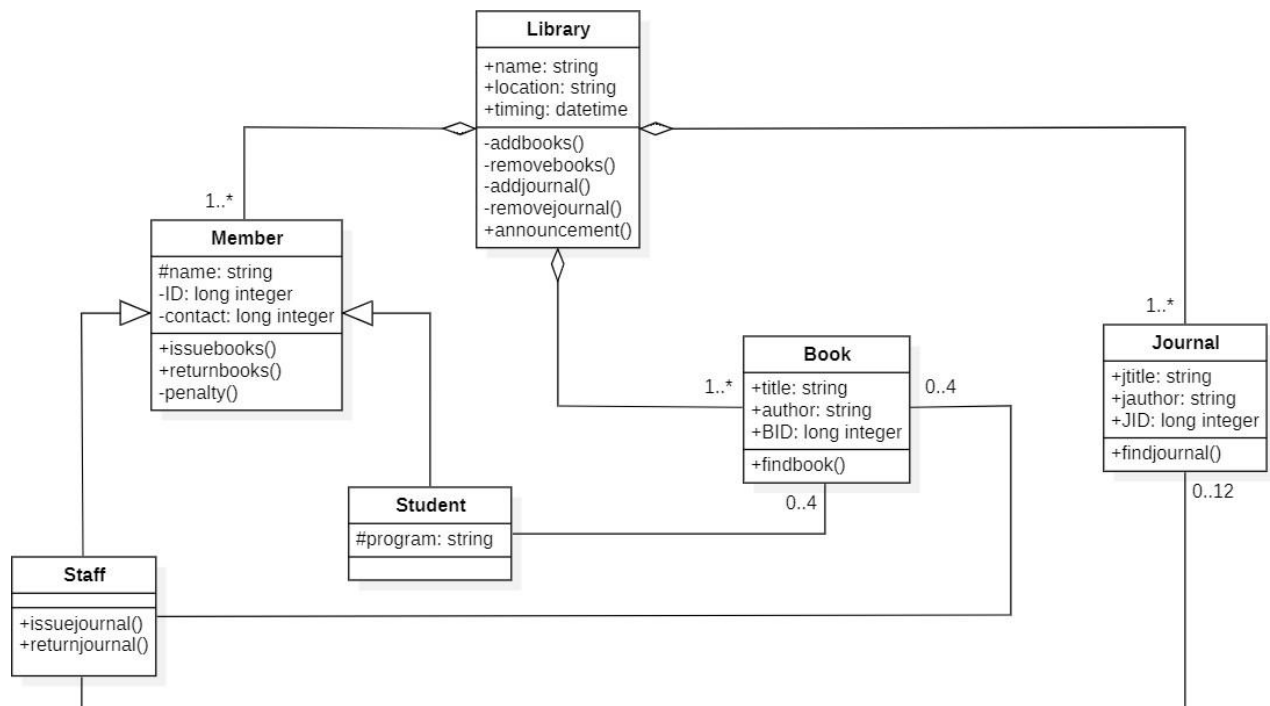


2. Draw class diagram of Library Management system:

Hint: classes are Library, Book, Journal and Members. The central class is library, Associated with Members who can issue books or journals depending upon one of two types of member's student or staff. Staff has privilege over student to issue journal(s) as well while student can only issue book(s). Library is also associated to Book and Journal in such a way that each book/journal only associated to exactly one library. Staff can issues maximum of 12 books per semester but students can issue maximum of 6 books per semester. Attached Printout here

Solution: For question 2, we assumed the following:

- We assumed location and timings for the library class.
- We also assumed a method announcement in the library class. This method will notify about announcements like public holidays, etc.
- In the Member class, we assumed a method penalty which is for if the book/journal is not returned on the due date.



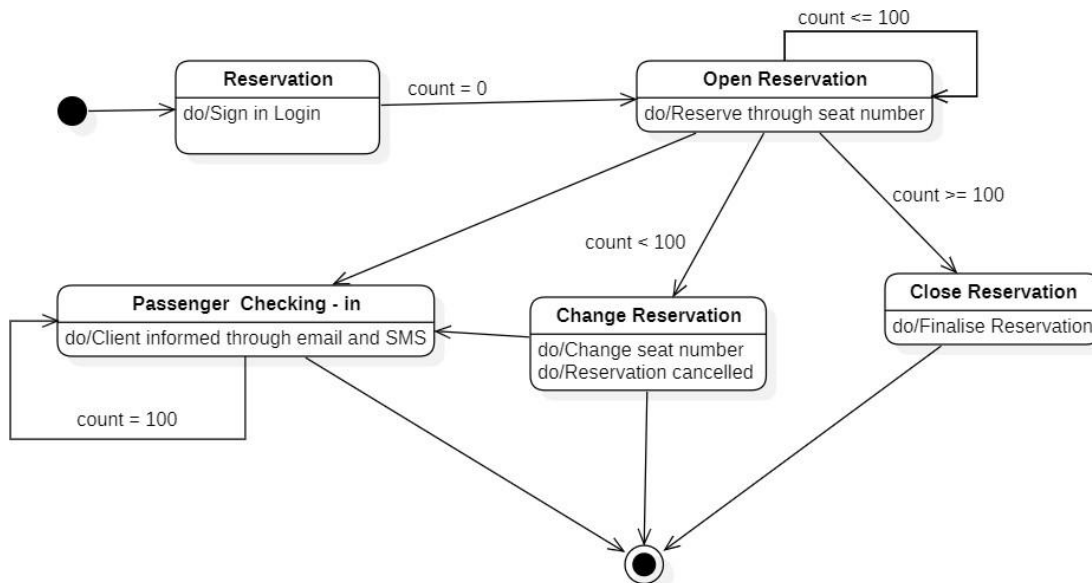
LAB SESSION 5

1. Draw State Diagram for Airline Reservation System.

Show different activities involved for making Reservation, Changing Reservation and Passenger Checking in for flight. Also clearly write down your assumptions.

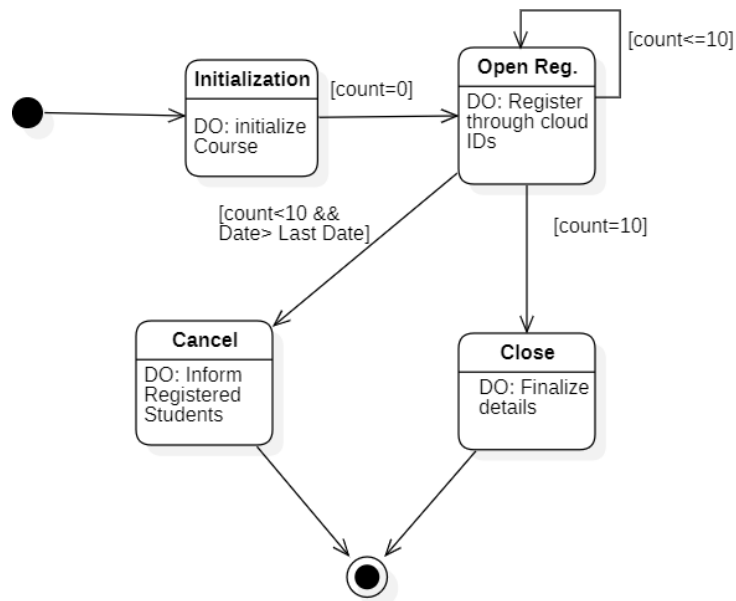
Solution: For this diagram we assumed the following:

- All users will receive confirmation email and SMS after successfully booking their flight.
- Assuming that the maximum capacity of the flight is 100, the option to change reservations will be closed once the count reaches 100.
- Users will not be able to make any changes to their reservations once the plane capacity reaches 100.



2. Draw State Diagram of University for course registration system

Details: NED University offers new course; course can initiate only if 10 students (exactly) registered in course. Registration will cancel otherwise; students must have to login with their cloudIDs for registration.

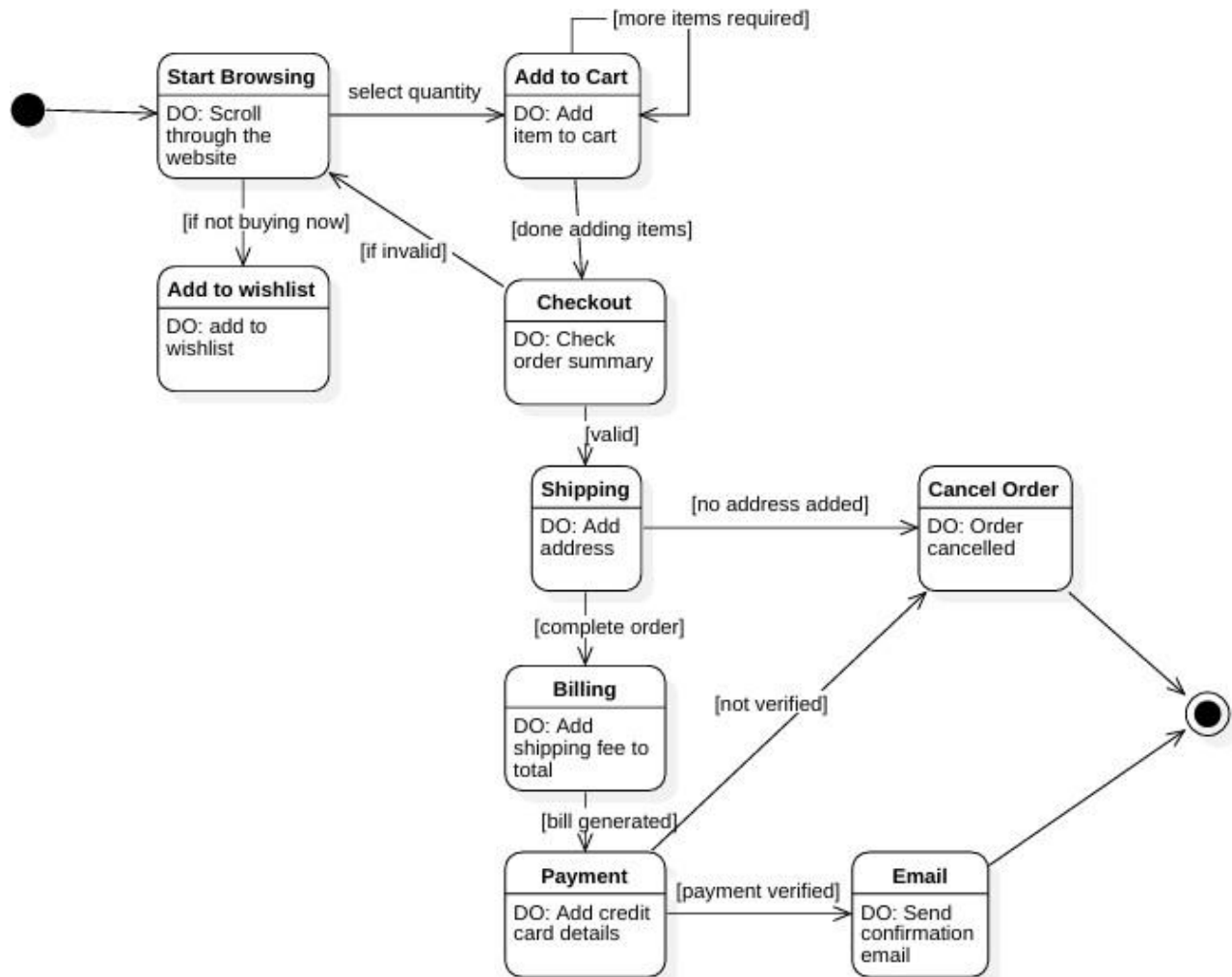


3. Draw State Diagram of online shopping system.

Consider any website and define states and their transition to successfully place order, also define appropriate start and exit states. Also clearly write down your assumptions

Solution: For this diagram we had the following assumptions:

- The user does not need to sign up or sign in to place an order.
- The checkout state being invalid means that items are either missing or in the wrong quantity.
- Without shipping information, the order cannot be processed.



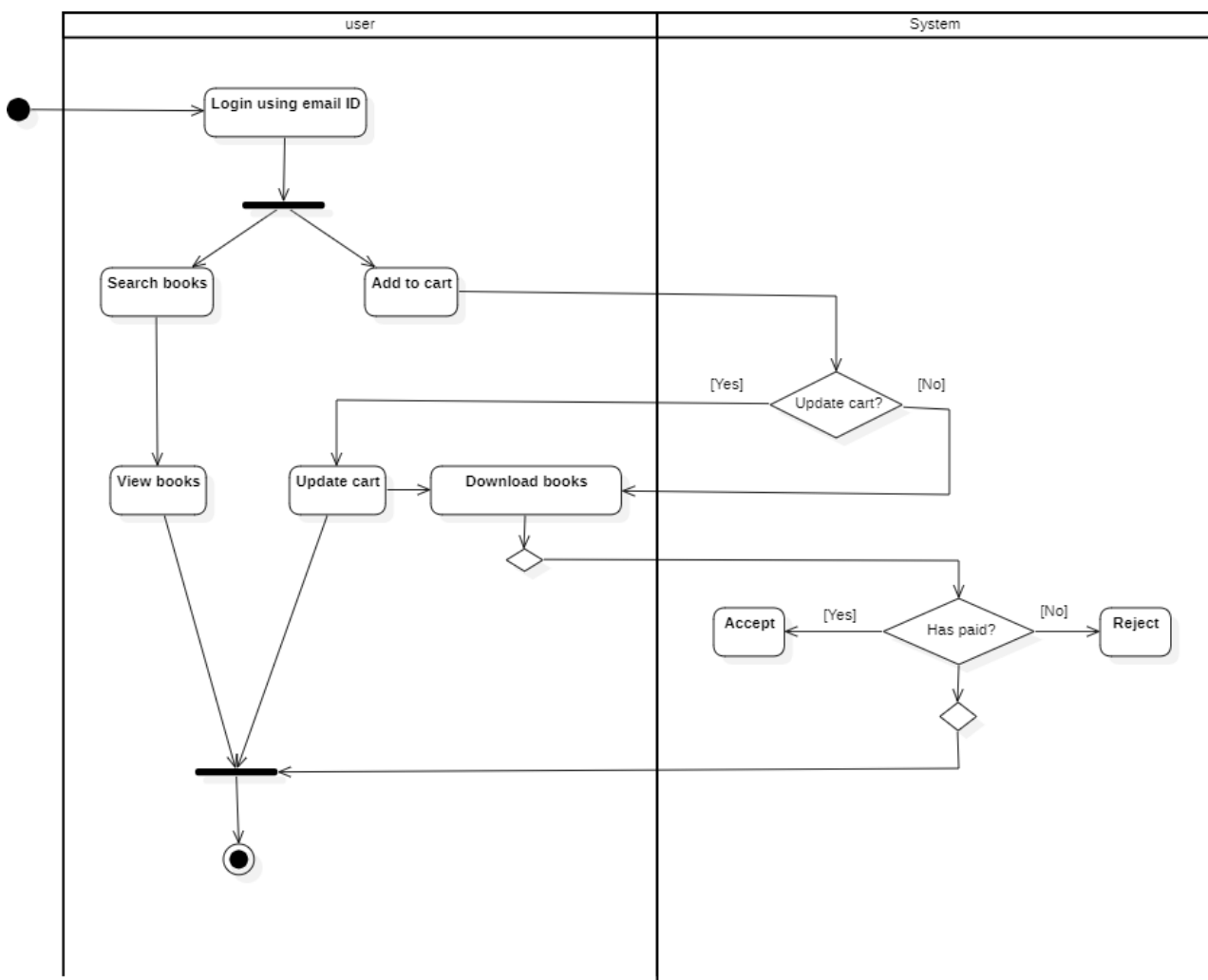
LAB SESSION 6

1. Draw Activity Diagram of the following:

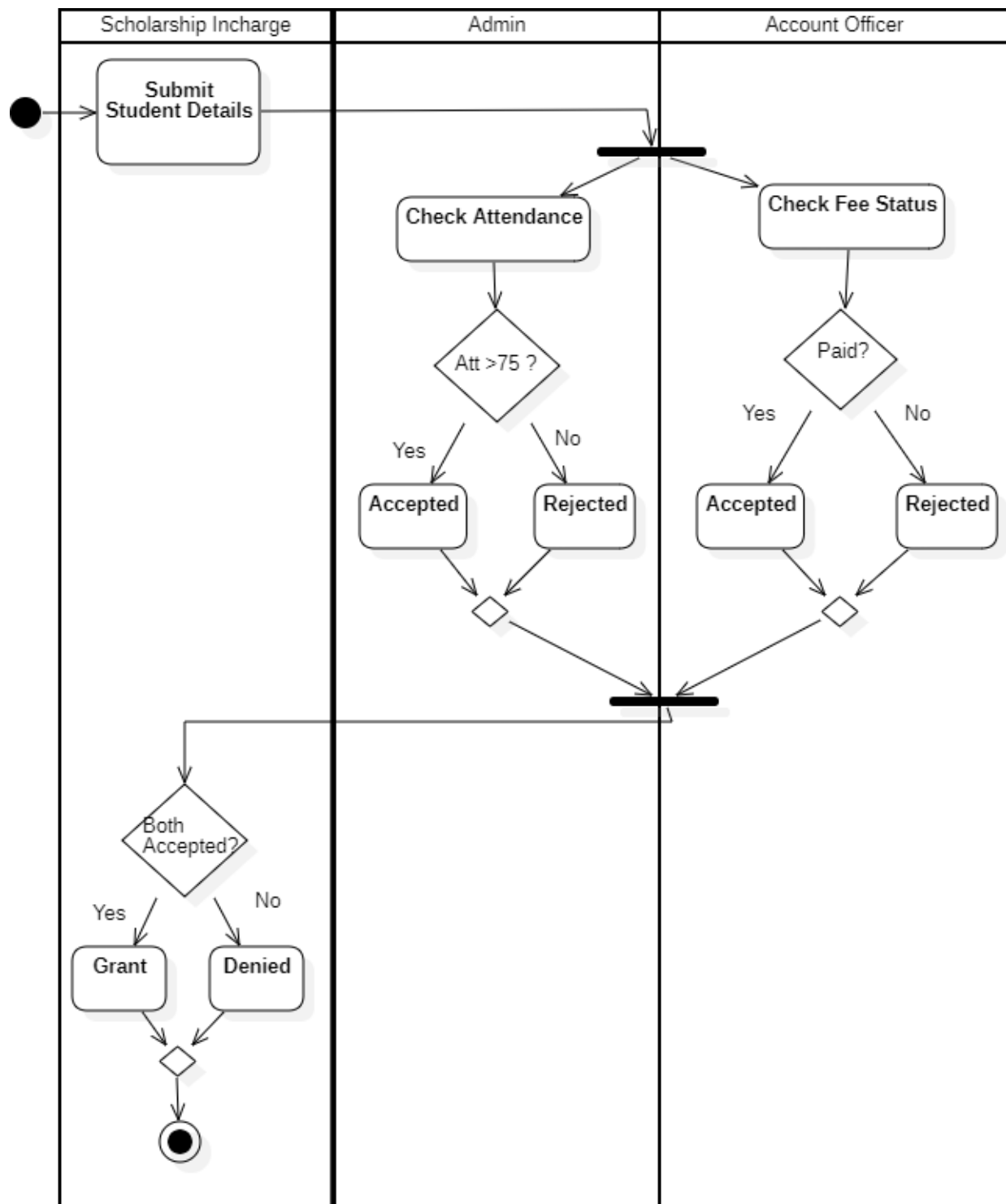
Suppose user can search books and view books online without paying any fee, but he can download book only if he pays, provided the facility to managing cart by using update feature. Make your own assumptions to complete the scenario.

Solution: For this diagram we assumed the following:

- The user may choose to update the cart. If they do, the cart is updated; if not, they can proceed to download books.



2. Draw Activity Diagram of scholarship approval system, make all necessary assumptions.



3. Suppose we want to develop a web based social media network with the following functionalities:

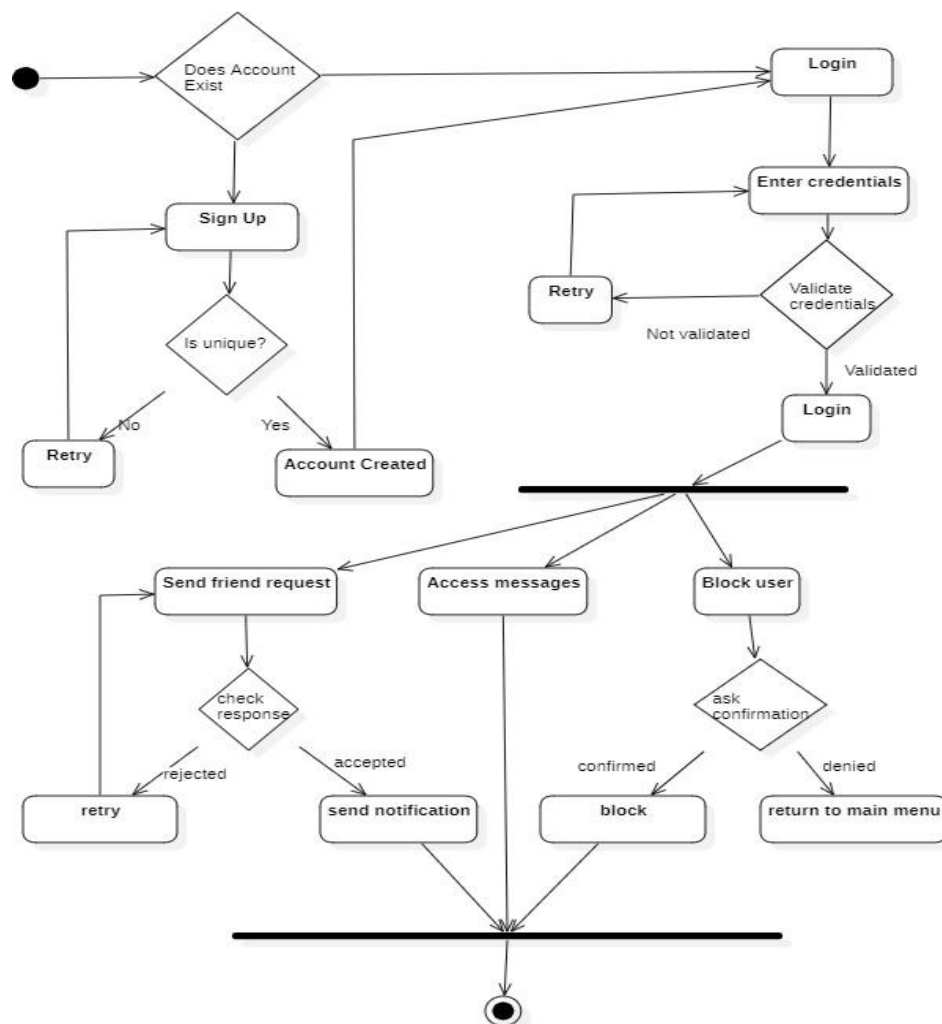
- The user should signup for the system
- The user should loginto the system and can send friend request or block the user

Write your assemptions against each activity and Draw activity diagram.

Hint: Activity1: signup, Activity 2: login, Activity3: send friend request, Activity4: Block user

Solution: For this diagram we assumed the following:

- We assumed the user could do three things after logging in, one was to access messages, the other to send a friends request and the third to block a user.
- We assumed user credentials would need to be verified to create an account or login in to an existing account.
- We assumed notifications would be sent to the user if a friend request was accepted and if not they would be prompted to retry if needed.
- We assumed we would need unique credentials to create a new account.



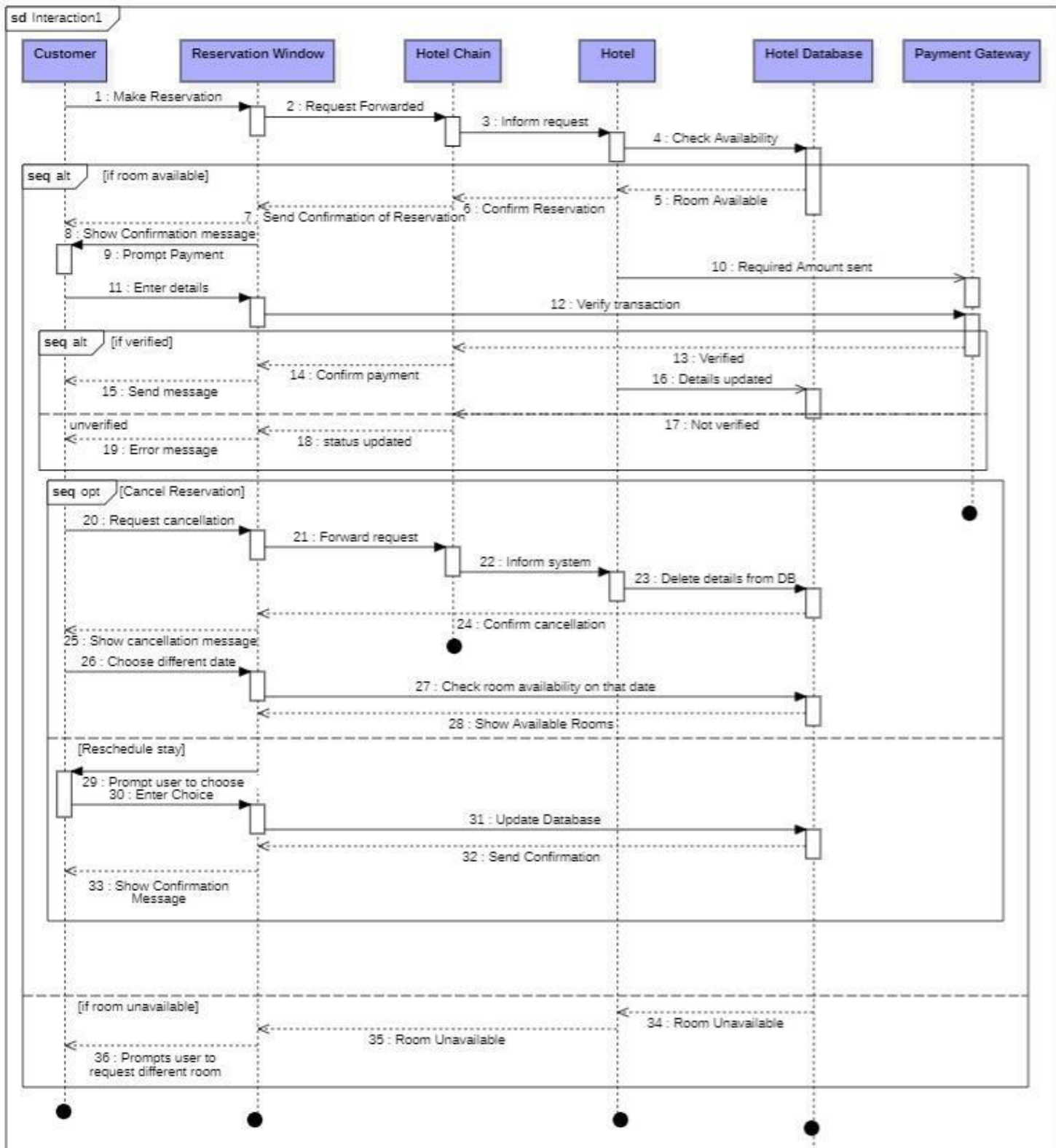
LAB SESSION 7

1. Draw a sequence diagram for Hotel Reservation System

Hint: The object initiating the sequence of messages is a Reservation window. The Reservationwindow sends a makeReservation() message to a HotelChain. The HotelChain then sends a makeReservation() message to a Hotel. If the Hotel has available rooms, then it makes a Reservation and a Confirmation. Explore the scenario and Write Down your assumptions also attach printout of diagram here.

Solution: We Assumed:

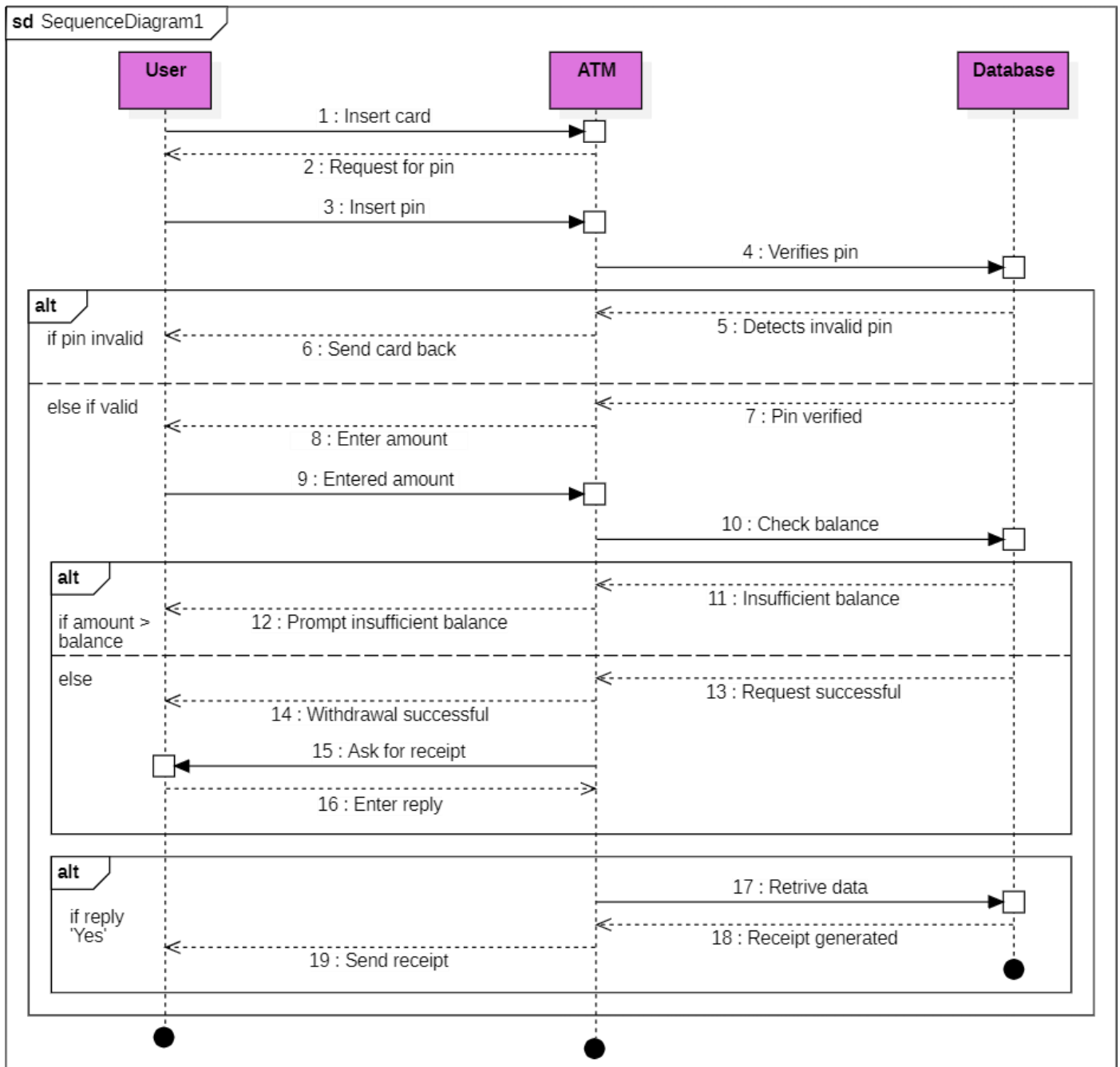
- The Hotel has a Database that responds to queries
- The User could choose between 2 options to edit their reservation if the room was available, to cancel their reservation or reschedule their reservation.
- Room availability would be verified to the user via the reservation window



2. Consider Cash withdrawal through ATM and construct sequence diagram. Explore the scenario and Write Down your assumptions also attach printout of diagram here

Assumption:

- User will insert a card and if pin is invalid, they will receive it back without withdrawing an amount.
- If the pin is valid, they have to enter an amount less than their current balance else the system prompts an error.
- After withdrawing an amount, they can also choose to generate a receipt.



LAB SESSION 8

TASKS

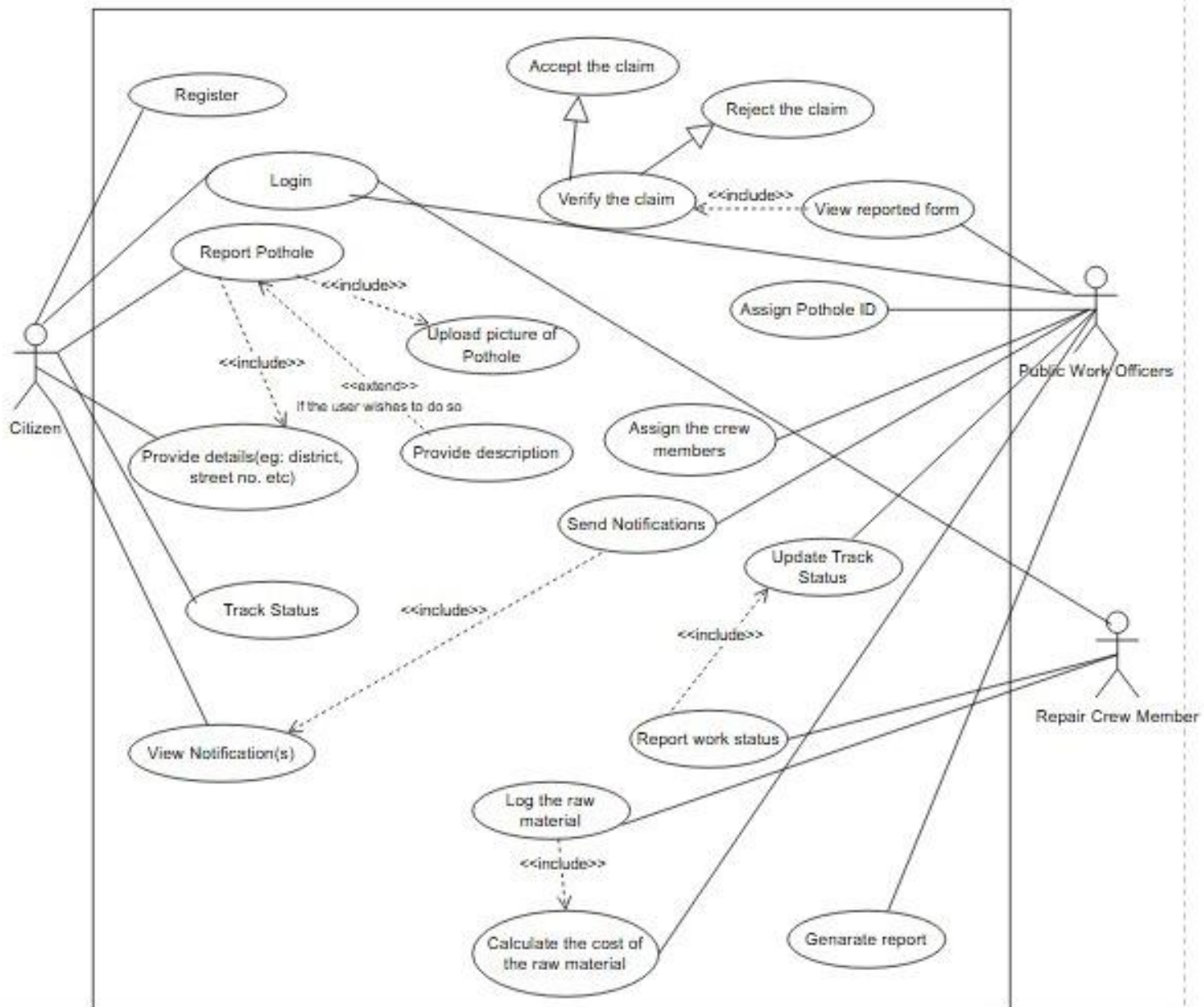
The department of public works for a large city has decided to develop a Web-based pothole tracking and repair system (PHTRS). A description follows: Citizens can log onto a website and report the location and severity of potholes. As potholes are reported they are logged within a “public works department repair system” and are assigned an identifying number, stored by street address, size (on a scale of 1 to 10), location (middle, curb, etc.), district (determined from street address), and repair priority (determined from the size of the pothole). Work order data are associated with each pothole and include pothole location and size, repair crew identifying number, number of people on crew, equipment assigned, hours applied to repair, hole status (work in progress, repaired, temporary repair, not repaired), amount of filler material used, and cost of repair (computed from hours applied, number of people, material and equipment used). Finally, a damage file is created to hold information about reported damage due to the pothole and includes citizen’s name, address, phone number, type of damage, and dollar amount of damage. PHTRS is an online system; all queries are to be made inter- actively.

a. Write down number of assumptions about the manner in which a user interacts with this system.

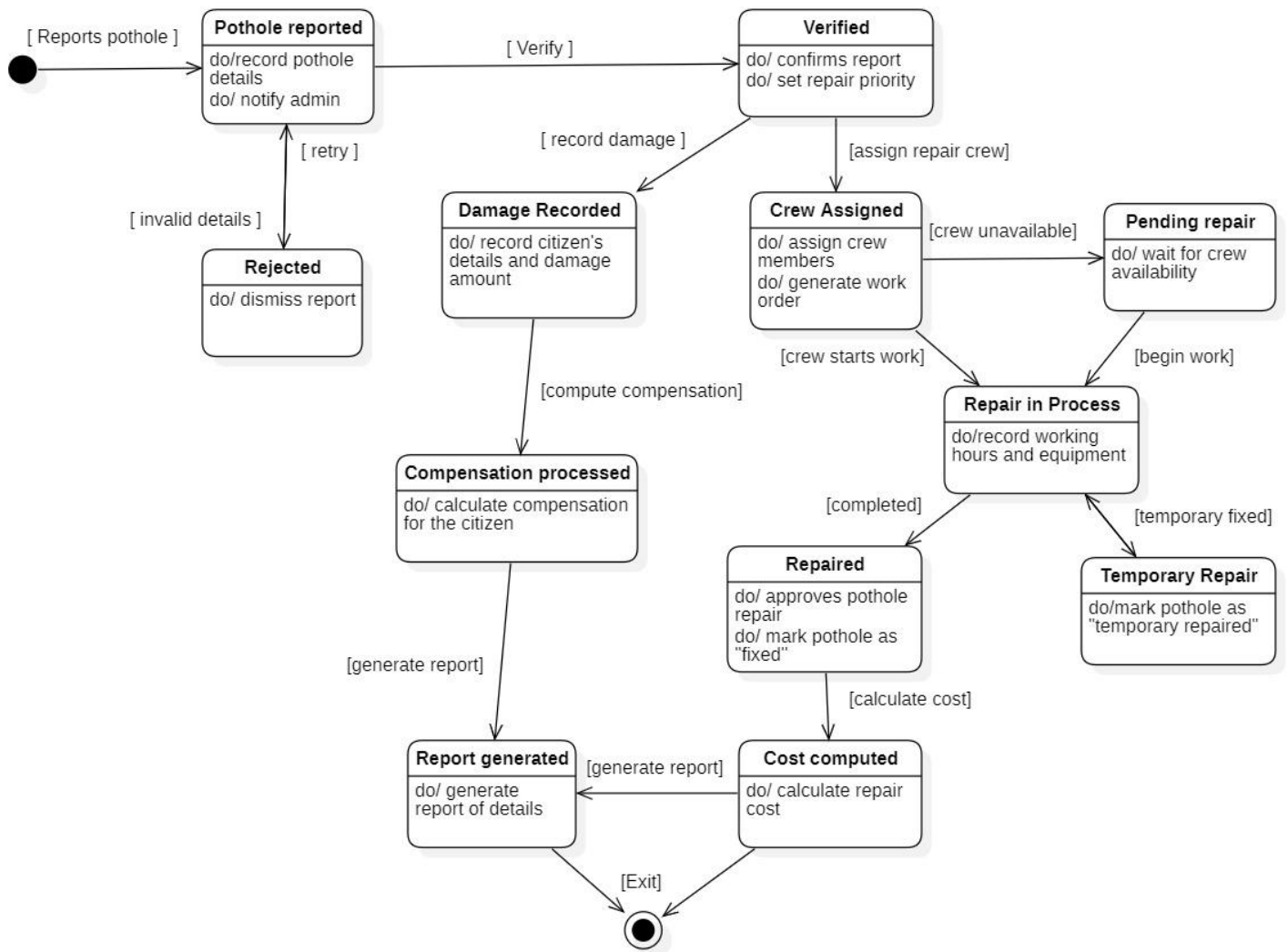
Assumptions:

1. Whenever a pothole is reported, the public work officer verifies it.
2. The officer either rejects or accepts the claim
3. The citizen has to upload the image of the pothole as well.
4. The citizen has a choice to give the description of the damage.
5. The public work officer also sends the notifications regarding the updates about the progress.
6. The citizen can view the progress/notifications.
7. There are three main entities who have access to the System, namely; Citizens, Repair Crew members and Public Works Officers.
8. The citizen has 5 main functionalities, out of which Login and registration are two. The others are reporting a pothole in an area, which includes certain elements.
9. Repair crew members are assigned by the public works officers.
10. Public works officers assign pothole ID’s to accepted potholes and update track progress.

b. Draw a UML use case diagram for the PHTRS system. (Attach Printout here)



c. Develop a State Chart Diagram for the PHTRS system (Attach Printout here)



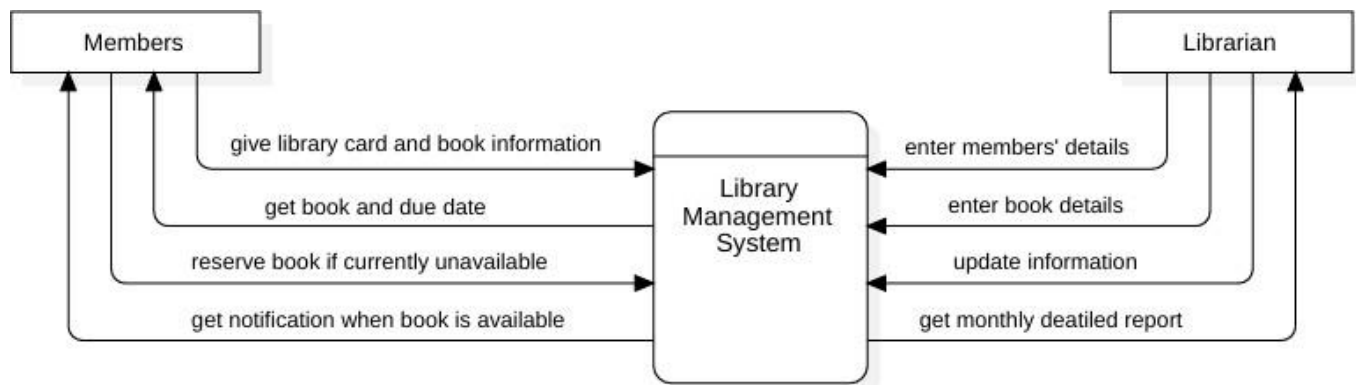
LAB SESSION 9

1. Draw Context Diagram and Level 1 Diagram of Library Management System, write down requirements and attach printout here.

Level 0

Solution:

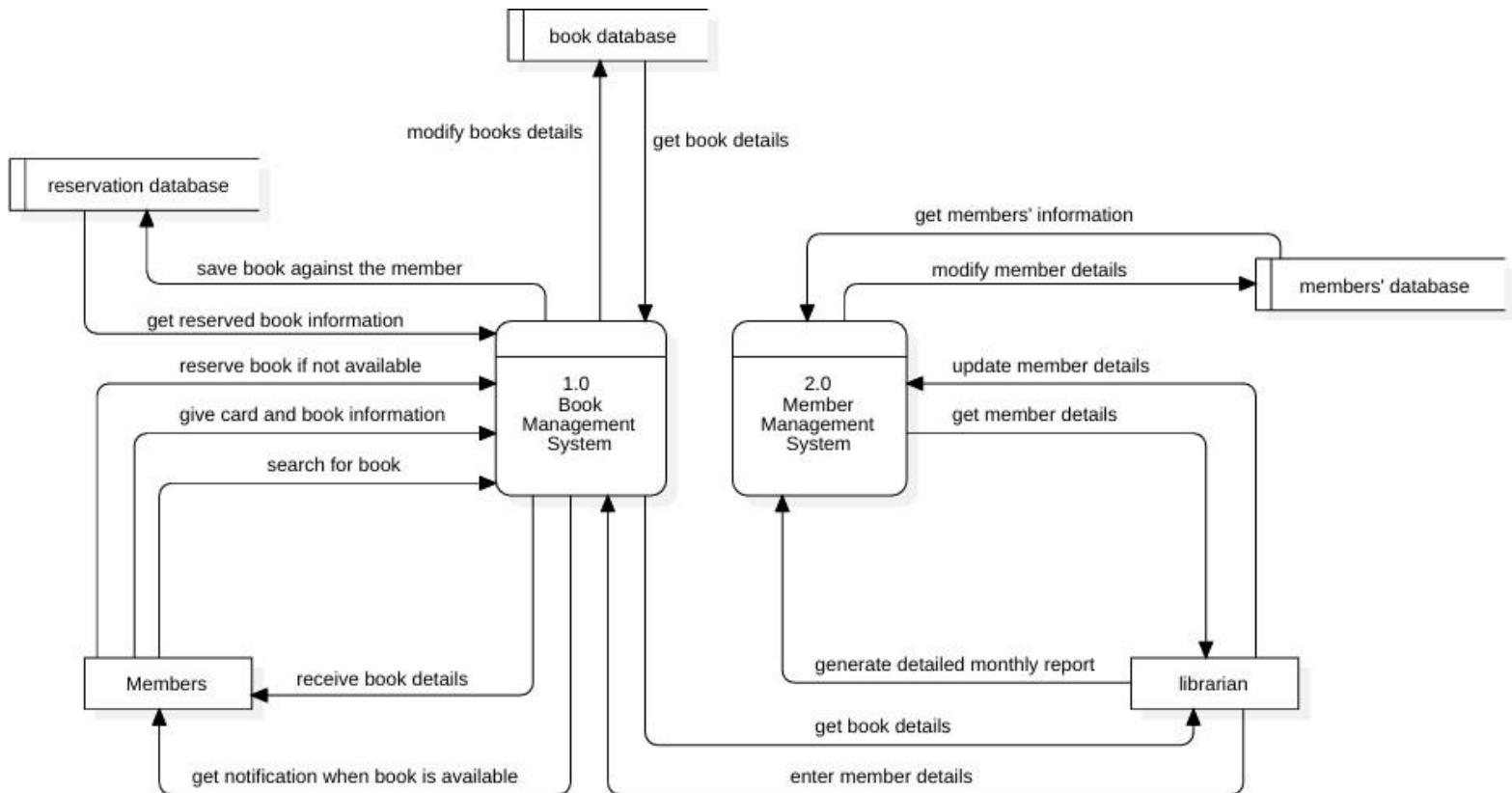
- Members can use their library card and enter book information to borrow books.
- They will receive book and the date within which they have to return.
- They can also reserve books if they are currently unavailable.
- Librarian can enter member details and issue library card.
- They can enter books details and generate a monthly report.



Level 1

Solution:

- Members can search for books.
- They can then provide their library card and the details of the books they want to borrow.
- If the book is currently unavailable, they can reserve the book and receive a notification when it is available.
- Librarian can view and update Member and book information.
- They can retrieve information from both the book and member database in order to generate and monthly report.



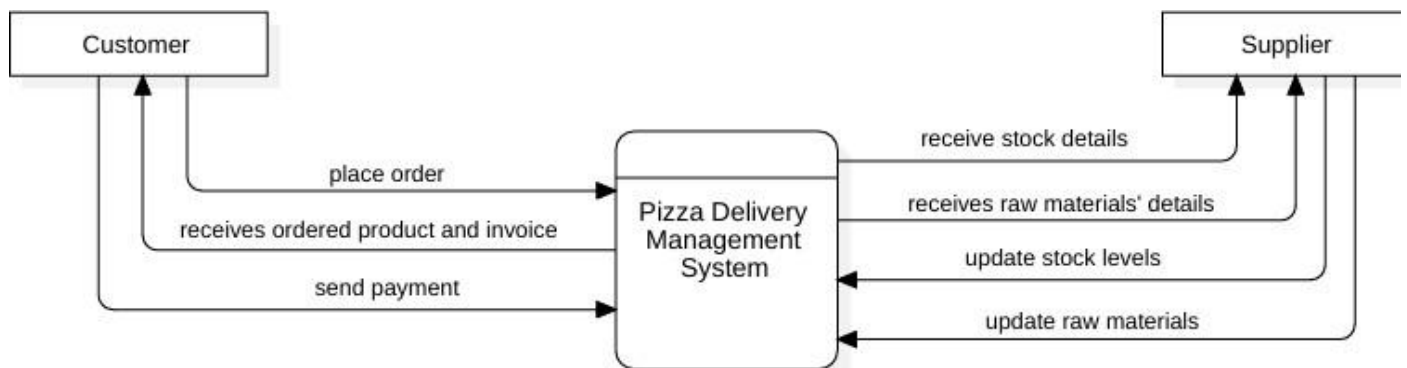
2. Case Study – Pizza Supreme

A large pizza business makes pizzas and sells them. The pizzas are manufactured and kept in cold storage for not more than two weeks. The business is split into a number of functional units. There is Production Control, Manufacturing, Stores, Accounts, Sales, Shipping and Purchasing. Production Control is responsible for organizing which pizzas to produce in what order and in what quantity. They need to schedule the production of the pizzas according to the current and expected sales orders together with the number of pizzas already in Stores. Manufacturing take the raw materials from the Stores and manufacture pizzas returning the completed goods to the Stores. Accounts deal with the payments for the pizzas when delivered to the customer and the payment to the suppliers of the raw materials. Sales deal with customer orders whilst Purchasing organize the buying of raw material from suppliers. Shipping manages the packing and delivery of the goods to the customer with a delivery note. When a sales order is received by sales they record what is being ordered and by whom. They also record the details of the expected date of delivery. Production Control access this information and make sure that, if required, pizzas are produced by Manufacturing and are ready in Stores for when the delivery needs to be made. After the delivery is made Accounts make sure that the customer receives an invoice and that payment for the invoice is received at which time a receipt is issued. Purchasing look at the current stock of raw materials and by using current stock levels, supplier turnaround times and quantity to be ordered decide what needs to be ordered on a daily basis. Their aim is never to run out of an ingredient but to minimize the amount of raw material kept in stock.(Attach Printout here)

Level 0

Solution:

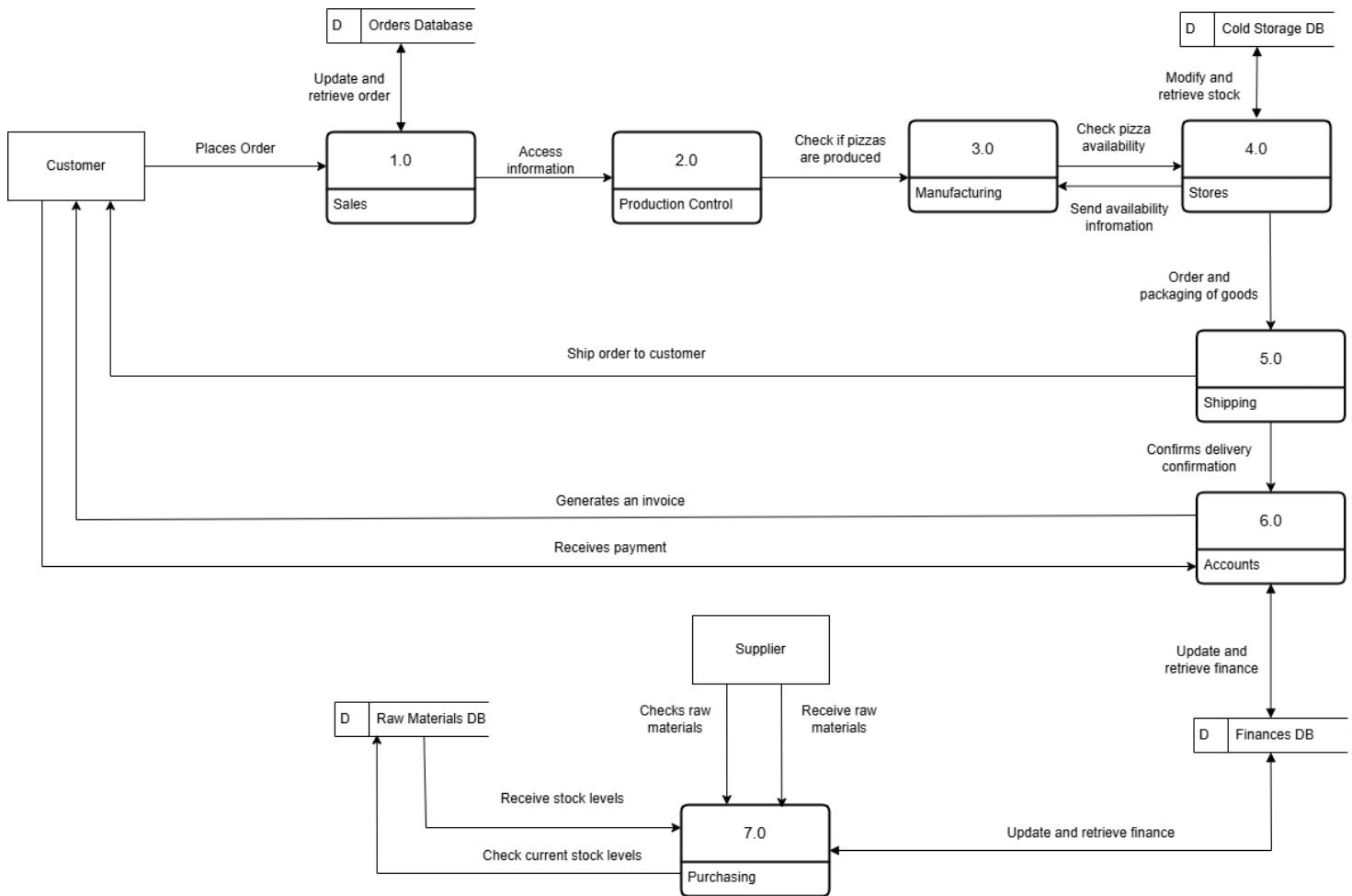
- Customers can place order and receive it along with their invoice.
- They can then provide the payment.
- Supplier will receive stock levels along with the amount of raw materials present.
- They can also update the information as deemed necessary.



Level 1

Solution:

- The Accounts process generates an invoice for the customer.
- The customer can then provide payment which is saved in the Finances database.
- Since, it is the supplier's job to check if the company is running out of raw materials, they can view and update stocks.
- Supplier can then make purchases and therefore retrieve and update finances in the database.

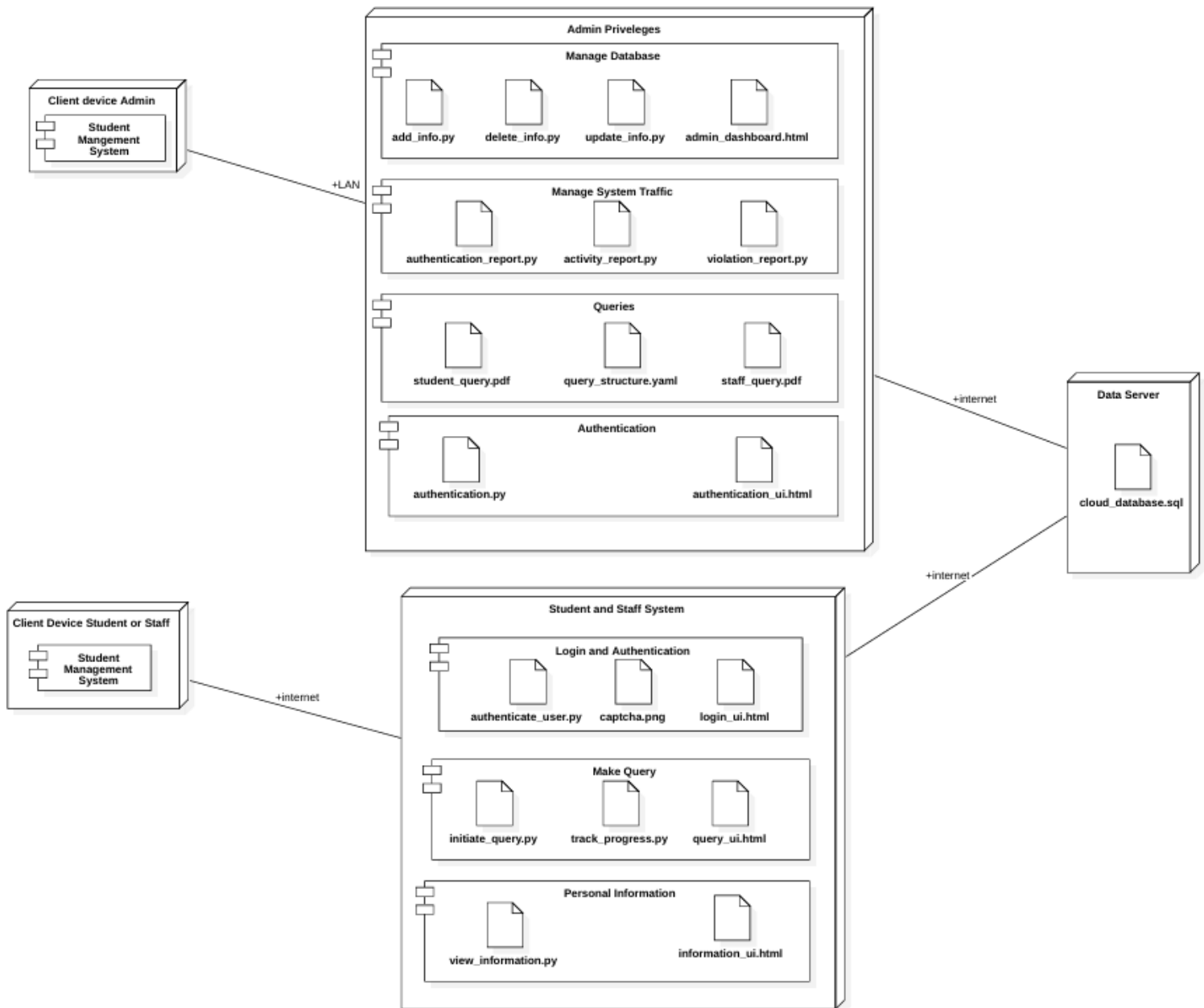


LAB SESSION 10

1. Draw deployment diagram of Student Management System:

Where Student /Staff /and admin can be the external nodes, admin can connect to system to manage and update database activities through private network (say LAN), and staff/students can interact with system through internet. Student database can store staff data file and student data file. An online database server will provide the customize view of student/staff's queries. Mention all required components with their attributes and possible functions inside management system. (Attach Printout Here)

Solution:

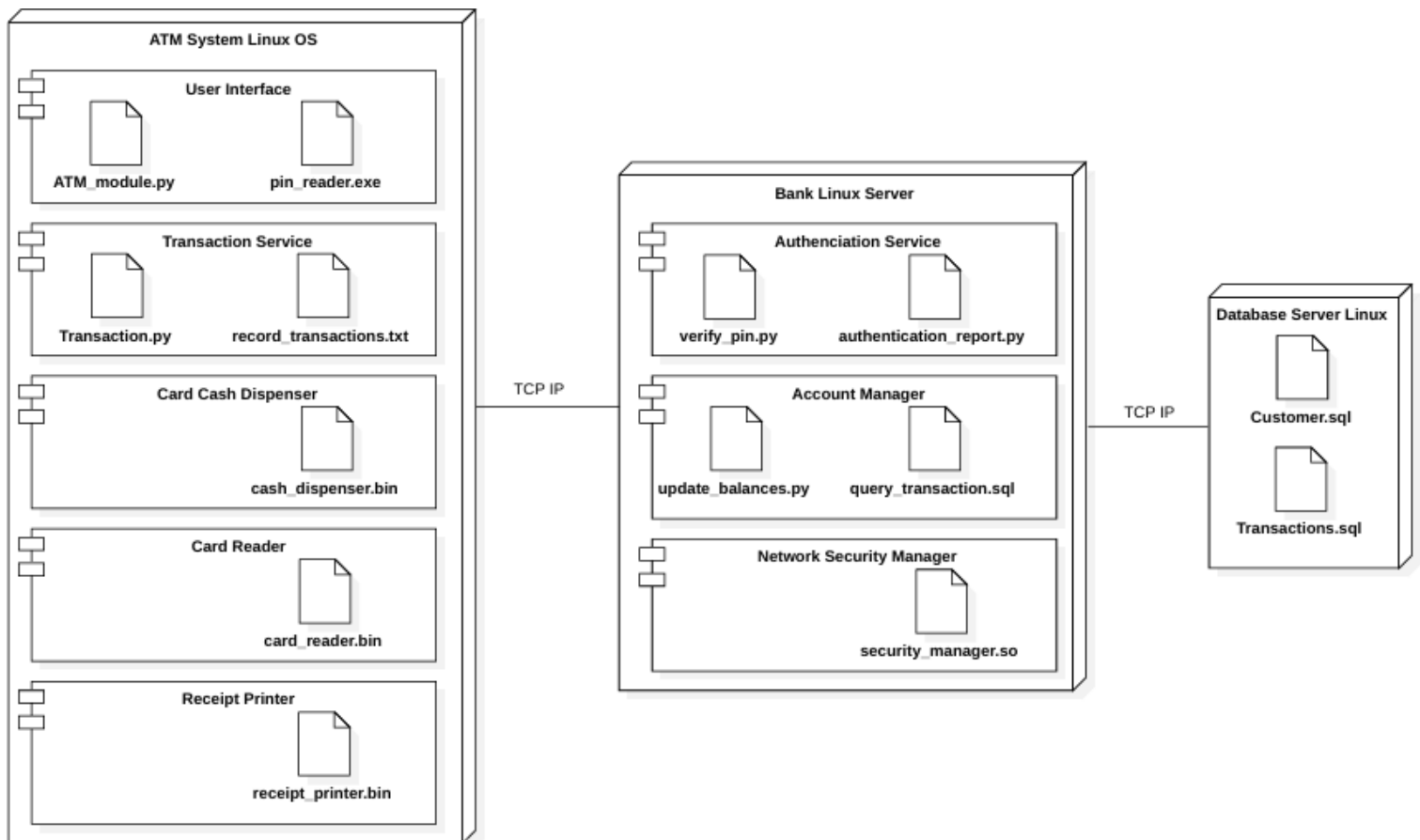


2. Draw deployment diagram of ATM, ATM software can only execute on Linux operating system,

1. add all possible components of software with attributes and operation, and show interaction with
2. devices involved in the system.

Solution:

- An ATM System is connected to the Bank Server which is then connected to the Database Server.
- The ATM handles user interface that reads the pin as well as other hardware components including card reader, cash dispenser and receipt printer.
- Customer transactions are also recorded.
- Bank Server manages the transactions and balances, authenticates services (including generate reports) and manages network security.
- The Database Server records the customer and their transactions.



LAB SESSION 11

Scenario:

1. A customer uses a mobile app to place a food order from a restaurant. The app sends the order details to the restaurant system, which confirms the order, prepares the food, and notifies the delivery service. The delivery service assigns a driver, picks up the food, and delivers it to the customer.

Instructions:

- a. Identify the objects/instances involved in this process (e.g., Customer, MobileApp, RestaurantSystem, DeliveryService, Driver).
 - Show the sequence of messages exchanged to complete the process:
 - Customer selects items and places an order via the Mobile App.
 - Mobile App sends the order to the Restaurant System.
 - Restaurant System confirms order and updates order status.
 - Restaurant System notifies Delivery Service.
 - Delivery Service assigns driver and updates status.
 - Driver picks up and delivers food to Customer.
- b. Use message numbering (1: placeOrder(), 2: sendOrderDetails()) to indicate the order of interactions.
- c. Include return messages where appropriate (e.g., orderConfirmation).

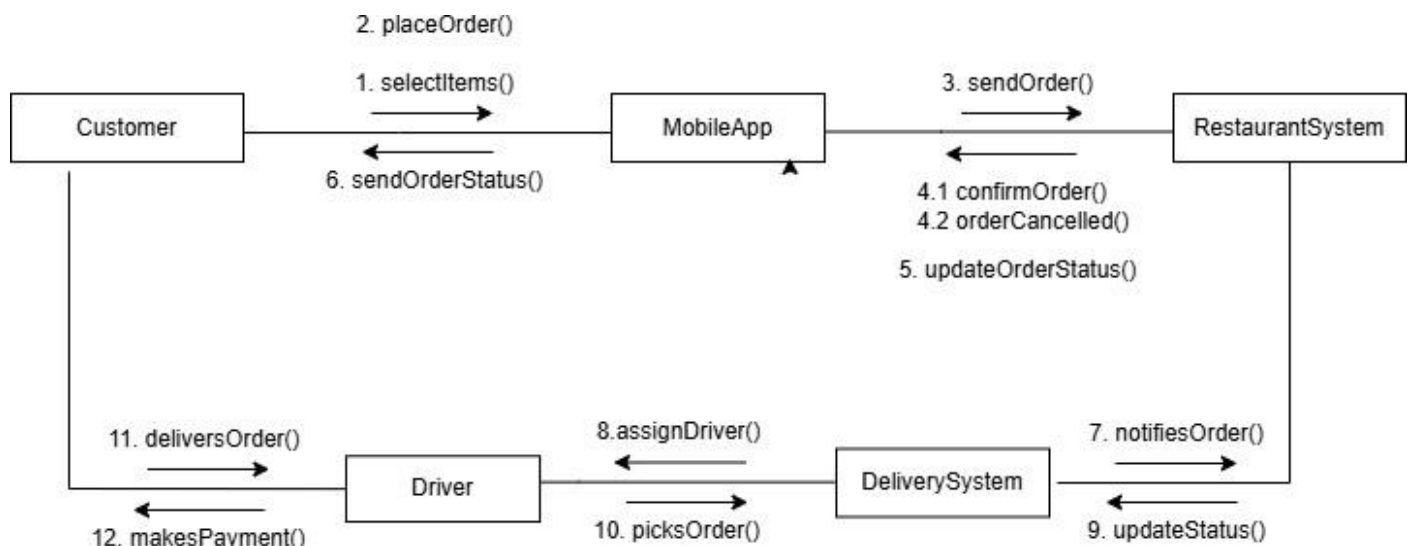
Deliverable:

A UML Communication Diagram that:

- Shows all involved objects.
- Correctly numbers messages in execution order.
- Clearly illustrates interactions between customer, systems, and services.

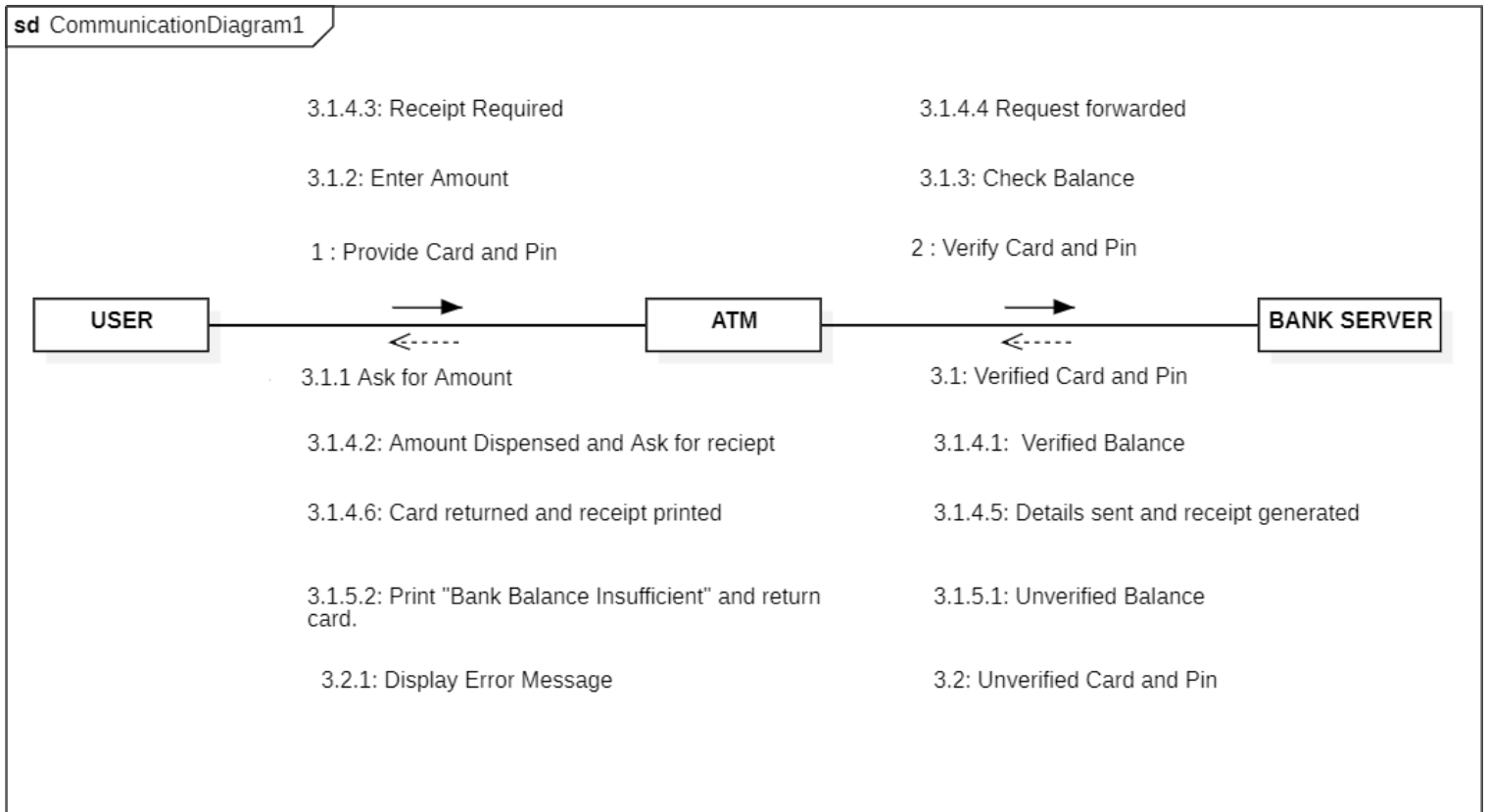
Solution:

- Customer will receive the status of their order from the Restaurant System that is if the order is confirmed or cancelled.
- The Delivery system will assign a Driver will receive the order and can update the order status once it is picked up by the Driver.
- The Driver will then deliver the order to the Customer who will then make the payment.



2. Draw communication Diagram of ATM Cash Withdrawal System.

Solution:

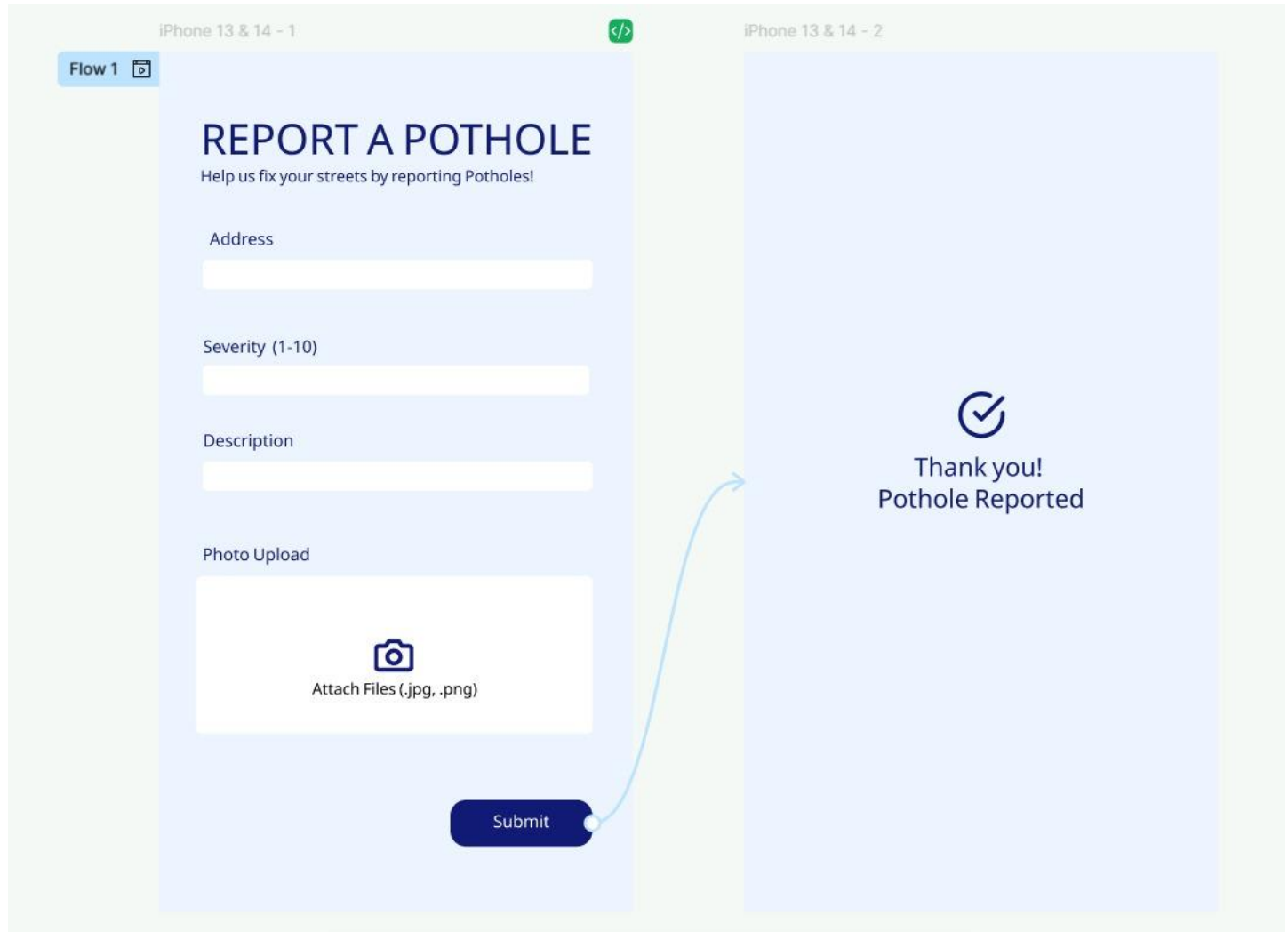


LAB SESSION 12

1. Build a simple “Report a Pothole” screen:

- Create a frame (mobile layout)
- Design input fields: Address, Severity (1–10), Description, Photo Upload, Submit Button
- Apply styling, align with layout grid, and convert Submit button into a component
- Prototype a submission interaction (Attached Printout here)

Solution:



LAB SESSION 13

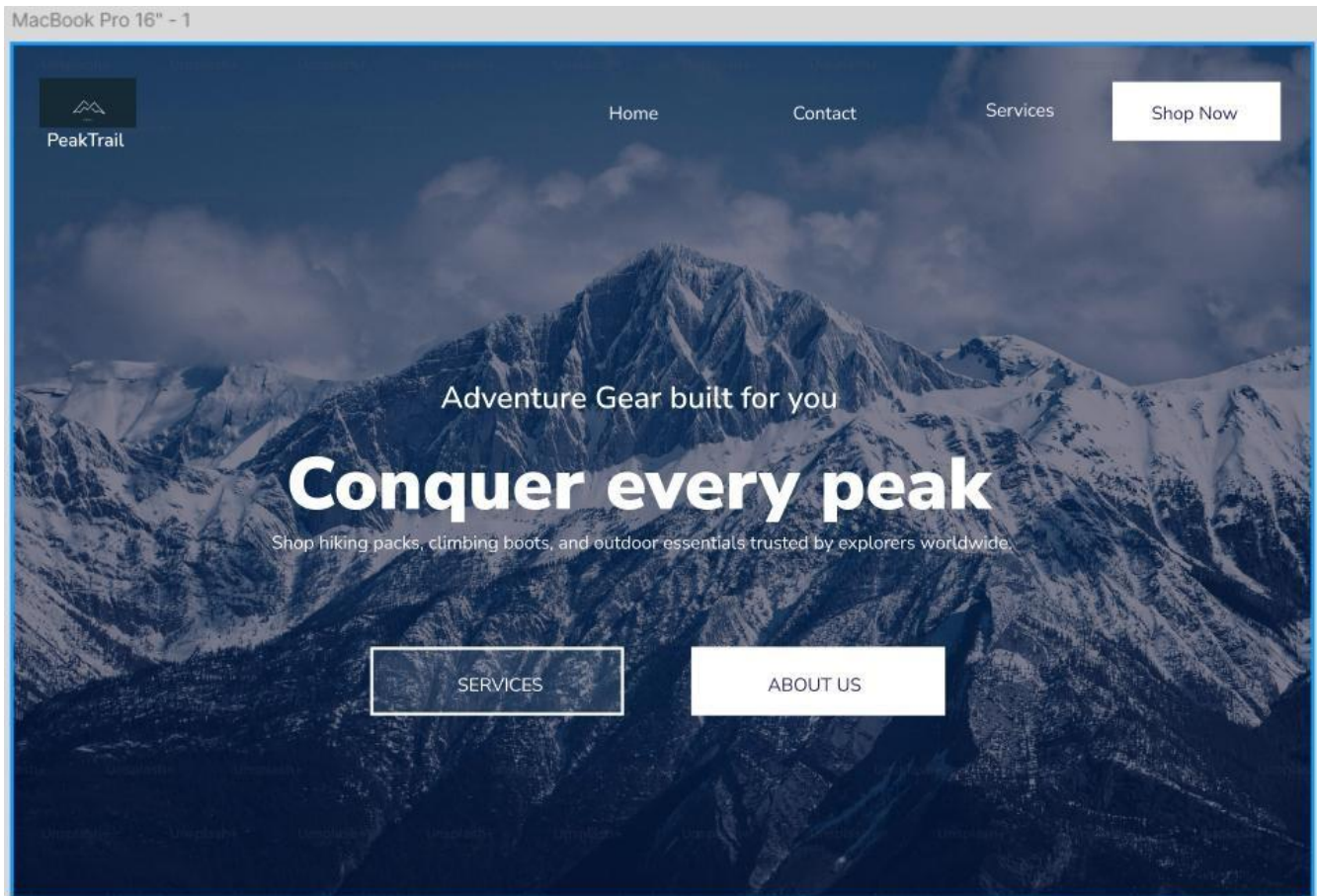
CEA – SUBMITTED WITH CEP ON GOOGLE CLASSROOM

LAB SESSION 14

1. Build high fidelity UI/UX design of Any E-Commerce Website and attach all printouts here.

We Created an E-Commerce Website for Equipment and Gear used by those who frequent the outdoors as campers, hikers and mountaineering professionals. The website contains a home page, a contacts page, an “about us” Information page about the vision and mission of the company and a Services page that redirects to the services and equipment sold on the website for each category.

Solution:



[Home](#)[Contact](#)[Services](#)[Shop Now](#)

About Us

At PeakTrail, we believe every adventure starts with the right gear. Our mission is to equip hikers, climbers, and explorers with dependable, sustainably made products that stand up to the toughest terrain.

From local trails to alpine peaks, we test every item we sell to ensure comfort, safety, and performance. Whether you're chasing your first summit or your hundredth, we're here to help you reach it.

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numbers

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- ✉ support@example.com

business hours

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 Saturday — 10am – 3pm
 Sunday — Closed



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Built for endurance and comfort on any trail. Our hiking collection combines lightweight materials, ergonomic design, and weather-tested durability to keep you moving. Whether it's a local trail or a multi-day trek, these essentials deliver performance you can rely on.



PeakStride Hiking Boots - \$129.00

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HydraFlow Flask 1L - \$32.00

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Designed for comfort and reliability under the open sky. From all-weather tents to compact cooking systems, our camping gear makes every trip efficient and enjoyable. Engineered for stability, ease of use, and long-lasting performance in any terrain.



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Pathfinder GPS Watch - \$245.00

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Precision-built for experts who work above the clouds. Every component in this line meets advanced safety standards and is tested in extreme alpine conditions. Trusted by professionals worldwide for strength, control, and absolute dependability when it matters most.



PeakPro Backpack - \$699.00

[Add to cart](#)

LunaBeam Lantern Camera - \$1749.00

[Add to cart](#)[See More -->](#)

Here is the prototyping, unfortunately it cannot be demonstrated through screenshots but the connections are visible on the screen here.

